0.00101, whereas likelihoods in the neighbourhood from 0.5 to 0.564 are higher than either of these.

Introducing another example, we will now focus on scenarios in which we take a sample of size n from a normal distribution N (μ, σ^2) population with known variance σ^2, testing whether or not the null hypothesis $H_0 : \mu = \theta$ on the mean holds (against the alternative $H_1 : \mu \neq \theta$). Without loss of generality, we shall set $\sigma^2 = 1$ and we also set $\theta = 0$. The Bayesian framework requires specifying what is meant by H_1, and for this we will examine more than one version. The first of these is Robert's (2014) Bayes factor for the null against the alternative when $\sigma^2 = 1$:

$$R\left(\bar{X}, n\right) = \sqrt{n+1}\exp\left(-\frac{n^2\bar{X}^2}{2n+2}\right) \tag{4}$$

We can readily confirm the claim by Robert that this equation yields $R(1.96/\sqrt{16,818}, 16,818) \approx 19$ in favour of the null. This reproduces the figures from Lindley's (1957) example, where he shows that a t-value of 1.96, sufficient for frequentists to reject the null at $\alpha = .05$, also yields a BF of 19 favouring the null. A bit of algebra reveals that to achieve a desired Robert BF of q we require

$$\bar{X} = \frac{\sqrt{(n+1)(\log(n+1) - 2\log(q))}}{n} \tag{5}$$

Two other versions of H_1 considered in this paper are from the JZS-scaled and information-scaled Bayes factors generated by the Rouder et al. (2009) t-test. The Robert BF is handy for its tractability, and the Rouder et al. BFs are included here mainly for the purpose of demonstrating the fact that our results generalize to some popular null-hypothesis BFs.

We now move to examining the behaviour of the Bayes factor when the null is compared against a point-valued alternative. A pointwise comparison between two alternatives gives a BF of

$$B\left(\mu_1,\mu_2,\bar{X},n\right)=\exp\left[n\left((\bar{X}-\mu_2)^2-(\bar{X}-\mu_1)^2\right)/2\right] \tag{6}$$

The log of this is linear in *n* and the rate of change in *n* is positive when $\left|\bar{X}-\mu_2\right|>\left|\bar{X}-\mu_1\right|$ and negative when $\left|\bar{X}-\mu_2\right|<\left|\bar{X}-\mu_1\right|$. Setting $\mu_1=0,$ to achieve $\log\left(B\left(0,\mu,\bar{X},n\right)\right)=\log\left(q\right)$, again straightforward algebra gives two solutions:

$$\mu=\bar{X}\pm\frac{\sqrt{n\bar{X}^2-2\log\left(q\right)}}{\sqrt{n}} \tag{7}$$

Given $\bar{X}>0$, say, for any μ such that $0 < \mu < 2\bar{X}$, $B\left(0,\mu,\bar{X},n\right) > 1$. By contrast, for any μ < 0 or $> 2\bar{X}$, $B\left(0,\mu,\bar{X},n\right) < 1$. Given any criterial threshold, *q*, from equation (7) we can find the sets of μ satisfying $B\left(0,\mu,\bar{X},n\right) \leq 1/q$ and $B\left(0,\mu,\bar{X},n\right) \geq q$. Moreover, we can use equation (5) to find $\bar{X}$ to input into equation (7) such that $R\left(\bar{X},n\right)\geq q$. We now have the wherewithal to ascertain the conditions under which Lindley cases occur (i.e., combinations of *q*, *n*, and $\bar{X}$ where $B\left(0,\mu,\bar{X},n\right) \geq q$ and $R\left(\bar{X},n\right)\geq q$). We will focus on large samples, because the impact of large samples seems to have been under-explored.

Bayes Factor Example

Suppose $n = 5{,}000$, and our BF threshold is $q = 3$. Then from equation (2) we have $\bar{X}=0.035557$. Suppose also that our BF threshold for *q* is 3. Inputting $n = 5000$, $q = 3$ and $\bar{X}=0.035557$ into equation (7) yields lower and upper bounds $\mu\in\left[0.0068368, 0.0642769\right]$. That is, $B\left(0,\mu,\bar{X},n\right) > 3$ against the null when when $0.0068368 < \mu < 0.0642769$. Figure 1 displays a graph of $\log\left(B\left(0,\mu,\bar{X},n\right)\right)$ as a function of μ, with two horizontal lines demarcating log(3) and log(1/3). The vertical lines mark out the two pairs of bounds that have just been derived.

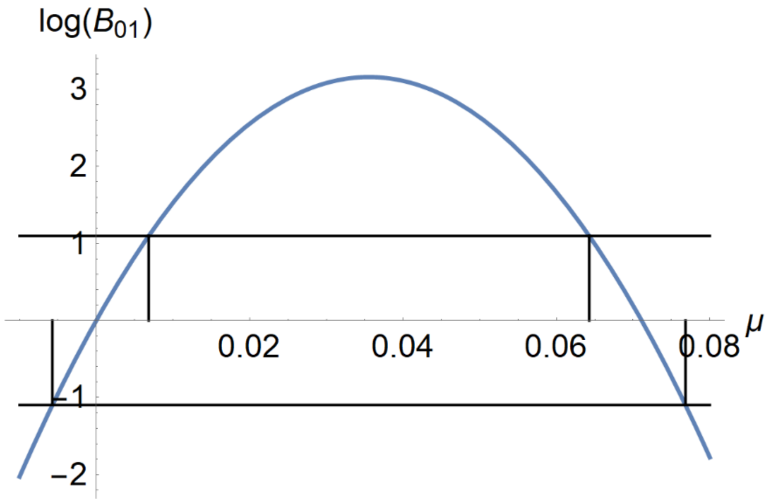

Figure 1. Log of Bayes Factor as a function of μ

In this example, we have seen that under a set of conditions for which the Robert (2014) BF $R(\bar{X},n) = 3$ in favour of the null hypothesis, there is a neighbourhood around $\bar{X}$ for which a comparison between the null and every alternative value of μ in this neighbourhood yields a BF greater than 3 against the null. How does the Rouder et al. (2009) Bayesian t-test fare here? The requirement that $\bar{X} = 0.035557$ to attain a BF threshold of 3 is equivalent to requiring a *t*-value of approximately $t = 0.035557\sqrt{5000} = 2.514$. Pursuing this with the JZS and scaled-information BFs from the Rouder test, it turns out that the *t* values required to obtain BFs ≥ 3 are 2.467 for the JZS and 2.373 for the scaled-information tests. Thus, the required sample means are $\bar{X} = 2.467/\sqrt{5000} = 0.03489$ for the JZS and $\bar{X} = 2.373/\sqrt{5000} = 0.03356$ for the scaled-information tests. These yield similar results to the preceding outcome using the Robert BF, i.e., a neighbourhood around $\bar{X}$ for which a comparison between the null and every alternative value of μ yields a Bayes factor greater than 3 against the null. These neighbourhoods are [0.00700, 0.06278] for the JSZ and [0.00735, 0.05977] for the information-scaled, both of which are fairly similar to the Robert-BF based interval derived above.